
Kaushik's KAC3 Model, Explained Simply

From First-Principles Statistics to a Full 5-Stock Worked Example

Written the way Richard Feynman would explain it -- if you can't explain it simply, you don't understand it well enough yet.

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Why this document exists: the first KAC3 methodology notes explained the model using its own vocabulary (factors, exposures, cross-sectional regression...). This one goes one level deeper: it builds up every underlying statistical idea from scratch, in plain language and everyday analogies, explains **why** the model has to be built the way it is, and then works through a complete, genuinely computed 5-stock example -- every number in Part 3 was actually calculated, not invented, and can be checked by hand.

Part 1	The Building Blocks	Statistics from first principles -- returns, wobbliness, factors, regression
Part 2	Why The Model Is Built This Way	The impossible task, the big trick, and the subtle double-counting trap
Part 3	A Real Worked Example	5 stocks, 10 weeks, every step computed and checkable

PART 1

The Building Blocks

Every statistical idea the model relies on, built up from scratch, one small step at a time.

1.1 What is a "return"?

If you had \$100 last week and you have \$103 today, you made a **return** of +3%. That's all a return is: how much richer or poorer something made you, as a percentage. If a stock's return this week is -2%, it means \$100 invested in it became \$98. Nothing more mysterious than that -- every single idea in this document is built out of nothing but these simple percentage changes, week after week, stock after stock.

1.2 Wobbliness: variance and volatility

Imagine two kids get their allowance changed every week. Kid 1's allowance always moves by about +/- \$1 -- steady, predictable. Kid 2's allowance sometimes jumps by +\$10 and sometimes crashes by -\$8 -- wild, unpredictable. Kid 2 is "wobblier."

Think of it like this: To measure wobbliness with a number, take each week's wobble (the change from what you'd expect), square it (multiply it by itself), and average those squares over many weeks. Squaring does two useful things: it makes every wobble positive (so a +\$10 week and a -\$10 week don't cancel each other out to zero), and it punishes big wobbles much more than small ones -- a \$10 wobble contributes 100, while a \$1 wobble only contributes 1.

That average-of-squared-wobbles is called **variance**. Its units are a bit awkward ("dollars squared"), so we usually take the square root of it to get back to normal units -- that's called **volatility** or **standard deviation**. In this document, "weekly volatility" always just means: how big are this thing's typical wobbles, in a normal percentage, per week.

1.3 Do two things wobble together? Covariance and correlation

Now imagine both kids get allowance from the same parent. When that parent is in a bad mood, BOTH kids' allowance drops the same week. Their wobbles move together, even though the two kids are different people.

Think of it like this: To measure "do these two things wobble together," take each week's wobble for thing A times that same week's wobble for thing B, and average that product over many weeks. If A and B tend to wobble up and down together, that product is usually positive, so the average is positive. If they tend to move in opposite directions, the average comes out negative. If they're unrelated, the positive and negative products roughly cancel out, and the average is close to zero.

That average is called **covariance**. It's hard to judge just by looking at the number whether 0.7 is "a lot" of togetherness, so we rescale it into **correlation**, which is always squeezed between -1 (perfectly opposite) and +1 (perfectly together), with 0 meaning unrelated. Correlation is just covariance divided by the two things' own volatilities.

1.4 Hidden shared forces: "factors"

Think of it like this: Picture a bathtub full of toy boats, and someone blowing across the water. Every single boat wobbles a little bit from that shared wind -- that's a force acting on ALL of them at once. But each boat ALSO gets its own little private bumps from bath toys bumping into it, which have nothing to do with the wind and are different for every boat.

Stocks work the same way. Some days, almost every stock moves together because of one shared force -- "the whole market is up today." Other times, there's a subtler shared force: "today, unloved cheap stocks did better than expensive ones," or "today, whatever's been going up recently kept going up." We call these shared, wind-like forces **factors** -- in this model, three of them: **Market**, **Value**, and **Momentum**. A stock's **exposure** to a factor is simply how strongly that particular boat catches that particular wind.

1.5 Finding the simplest honest explanation: regression

Think of it like this: Suppose you measure the height and shoe size of 100 kids. Taller kids tend to have bigger feet, but not perfectly -- there's scatter. **Regression** is just the process of drawing the single best straight line through that scatter of dots: the line that, on average, is as close as possible to everyone's actual shoe size. It won't be exactly right for any one kid, but it's the best simple rule you can write down.

The gap between a kid's actual shoe size and what the line predicted for their height is called the **residual** -- "the part that's just this one kid being their own kid," not explained by the shared height-to-shoe-size rule. In the stock model, this leftover part is called the **specific return**: whatever happened to a stock that isn't explained by Market, Value, or Momentum.

1.6 Comparing apples to apples: standardizing (z-scores)

You scored 78 on a test. Is that good? It depends entirely on what everyone else scored -- 78 is amazing if the class average was 55, and mediocre if the average was 92. A **z-score** answers "how many wobble-units above or below the average was this," by subtracting the average and dividing by the typical wobble (the standard deviation). A z-score of +2 always means "unusually high, about two typical wobbles above normal," whatever the underlying units were -- which lets you compare a stock's cheapness, its size, and its recent performance on one common ruler.

1.7 A snapshot vs a movie: cross-sectional vs time-series

A **cross-sectional** look is a single photograph of everyone at one moment -- all 5 stocks, one particular week. A **time-series** look is a movie of just ONE stock across many weeks. The KAC3 model's central trick, explained in Part 2, is that it takes a brand-new cross-sectional photo every single week, and then studies the resulting sequence of photos as if it were a movie -- getting the best of both.

PART 2

Why The Model Is Built This Way

The impossible task, the big trick that solves it, and a trap you have to watch for.

2.1 The impossible task

Think of it like this: Imagine someone asks you: "tell me exactly how all 3,000 kids in a giant school are connected to each other -- who's friends with whom, and how strongly." That's a huge number of pairs of kids. Now imagine you only get to watch the school for 5 days. There's simply no way to work out that many relationships from that little observation -- you don't have enough data to pin down that many unknowns.

This is exactly the problem with stock returns. A large universe might have several thousand stocks. Estimating how every single pair of stocks moves together directly (their raw covariance) needs about as many numbers as there are pairs -- millions of them -- but even years of daily data only gives you a few thousand days. Too many unknowns, not enough data. Any attempt to estimate that giant table directly comes out unreliable and, worse, can accidentally suggest "risk-free" combinations of risky stocks that don't really exist -- pure noise pretending to be a pattern.

2.2 The big trick: track a few shared winds, not every friendship

Think of it like this: Back to the school: instead of mapping every pair of kids, you notice that a lot of the grouping comes from a handful of shared things -- some kids like sports, some like music, some are in the same class. If you know how strongly each kid "leans into" a small number of these shared groups, you can explain a surprising amount of who-hangs-out-with-whom, using far fewer numbers than mapping every pair directly.

That's a **factor model**. Instead of estimating covariance between every pair of thousands of stocks directly, estimate: (a) how exposed each stock is to a small number of shared factors (Market, Value, Momentum), and (b) how those few factors wobble and move together. A stock's total risk then comes from combining those two much smaller, much more reliable pieces of information. This is the single most important idea in the whole model -- everything else is just making this idea work well in practice.

2.3 Taking a class photo every week: cross-sectional regression

Every single week, look at all the stocks' returns for that week, together with how exposed each stock already is to Market, Value, and Momentum (known ahead of time from each stock's own characteristics). Then find the best simple rule (Section 1.5's regression, but now run *across all stocks at one moment* rather than across one stock over time) explaining that week's moves from those exposures. The best-fit numbers that come out of that one week's photo **are** that week's Market, Value, and Momentum returns. Do this every week, and you get a whole movie of factor returns to study.

2.4 A clever recipe: pure factor portfolios

Think of it like this: If you want a pure, perfect red with none of your other paint colors mixed in, you figure out exactly which proportions of your paints, combined, cancel out every other color and leave only red. A **pure factor portfolio** is exactly that trick applied to stocks: a specific mix of holdings, long some stocks and short others, engineered so the mix has EXACTLY 1 unit of exposure to (say) Momentum and EXACTLY 0 exposure to Market and Value. Its return, week after week, then literally **is** that week's Momentum factor return -- not an estimate of it, but a portfolio you could actually hold whose return equals it.

2.5 Total wobble = shared wind wobble + everyone's own private wobble

Once you know (i) how wobbly each factor is and whether they wobble together (their covariance matrix, from Section 1.3 applied to the factor-return movie), and (ii) how wobbly each stock's own private leftover is (its specific variance, from Section 1.5's residuals), you can compute the risk of **any** portfolio, however it's assembled, without re-measuring anything from scratch:

```
total risk = risk from shared factors + risk from everyone's own private wobble
```

This is the payoff for all the machinery above: instead of needing a giant, unreliable table of every stock-pair's covariance, you only ever need a small factor covariance matrix (3x3 here) plus one specific-variance number per stock.

2.6 The subtle trap: measuring the ruler with the ruler

Think of it like this: Back to the bathtub: suppose you only have 2 or 3 boats, and you try to guess "how strong was the wind today" purely by watching those few boats. Some of what you see isn't really the wind -- it's those particular boats' own private bumps sneaking into your guess, because you don't have enough boats for the private bumps to average out and disappear. If you then ALSO separately add back "boats have their own private bumps" on top of your wind guess, you've counted some of the exact same bumps twice.

This is precisely what happens with pure factor portfolios: because they're built from a limited, finite number of stocks (never infinitely many), their observed return already contains a bit of genuine private/specific wobble mixed in with the true factor wobble. The traditional formula in Section 2.5 then adds a separate specific-risk term on top -- double-counting that same wobble. Part 3, section 3.8 catches this red-handed with real numbers.

PART 3

A Real Worked Example

5 fictional stocks, 10 weeks of returns, every single number computed and checkable.

3.1 Meet the five stocks

All five stocks always have exposure 1 to the Market factor (everyone catches the market's wind equally here, for simplicity). Value and Momentum exposures are chosen so that, across the five stocks, the average exposure to each is exactly zero -- meaning "the whole market together" is neither cheap nor expensive, neither hot nor cold, on net, which is exactly how the real model standardizes things (Section 1.6).

Stock	Character	Market	Value	Momentum
A	Big Steady Co. -- a large, boring, reliable business	1.00	-0.80	0.90
B	Cheap Value Co. -- unloved and inexpensive relative to earnings	1.00	1.50	-0.60
C	Hot Momentum Co. -- everyone's been chasing this one lately	1.00	0.30	1.20
D	Unloved Co. -- both pricey-ish AND out of favor right now	1.00	-0.50	-0.90
E	Middling Co. -- a bit expensive, mildly out of favor	1.00	-0.50	-0.60

Check: Value exposures sum to 0.00, Momentum exposures sum to 0.00 -- both exactly zero, confirming the market as a whole is style-neutral, as it should be.

3.2 Ten weeks of returns

Behind the scenes, I generated ten weeks of true (hidden) factor returns and each stock's own private noise, then combined them through the exposures above to produce the stock returns below -- so this data behaves exactly like the model assumes, and we can later check how well the model recovers the hidden truth. All figures are weekly percentage returns.

Week	A	B	C	D	E
1	1.31	1.19	0.02	-0.09	0.97
2	-1.39	-0.33	-2.79	-1.60	-0.63
3	2.58	1.19	1.67	0.66	0.82
4	1.16	3.15	1.93	2.06	1.45
5	-2.78	-1.43	-4.67	-2.53	-2.62
6	-1.75	-2.19	-0.05	-1.83	-0.56
7	-0.13	0.27	1.26	0.53	0.56
8	0.90	-1.28	-0.54	0.61	-0.19
9	-0.42	-0.06	-0.39	0.21	-0.06
10	-0.22	-1.44	-0.28	-0.63	-1.27

3.3 Taking the first "class photo": Week 1's regression, done by hand

For week 1, the regression finds the best-fit Market/Value/Momentum returns that explain all five stocks' moves that week simultaneously, given their exposures. The result that comes out is:

Week 1 recovered factors: Market = 0.678 Value = 0.089 Momentum = 0.009

Let's check this isn't magic by hand-verifying one single stock. Stock A's exposures are (Market=1.00, Value=-0.80, Momentum=0.90). Multiply each exposure by that week's recovered factor return and add them up -- this is exactly what "fitted return" means:

$$\text{fitted_A} = (0.678 \times 1.00) + (0.089 \times -0.80) + (0.009 \times 0.90) = 0.615$$

Stock A's actual week-1 return was 1.31%. The gap between what actually happened and what the factors predicted is the residual (Section 1.5): $1.31 - 0.615 = \mathbf{0.697}$ -- this matches the table in Section 3.6 exactly. That leftover 0.697 is "Stock A being its own stock" that week, nothing to do with Market, Value, or Momentum.

3.4 All ten weeks of recovered factor returns

Doing this same regression separately for every week produces a full movie of factor returns. The "true (hidden)" column shows what actually drove the data behind the scenes -- the model never sees this column, but we can compare against it here to see that the regression recovers something close to (though never exactly) the truth. It's never exact in any single week because five stocks and three factors leaves only two stocks' worth of leftover noise to average away -- exactly the "not enough data" tension from Section 2.1, just on a tiny scale.

Week	Rec. Market	Rec. Value	Rec. Momentum	True Market	True Value	True Momentum
1	0.678	0.089	0.009	0.727	0.440	-0.148
2	-1.348	0.164	-0.660	-1.156	0.389	-0.545
3	1.384	-0.050	0.659	1.351	0.033	0.978
4	1.949	0.717	-0.233	1.617	0.564	-0.124
5	-2.807	0.233	-0.867	-2.431	0.234	-0.343
6	-1.276	-0.140	0.422	-1.523	-0.430	-0.282
7	0.497	0.114	0.127	0.479	0.184	0.426
8	-0.101	-0.862	0.017	-0.143	-0.479	0.292
9	-0.142	-0.022	-0.258	0.276	0.439	0.330
10	-0.768	-0.261	0.403	-0.894	-0.025	0.345

3.5 How wobbly are Market/Value/Momentum, and do they move together?

Applying Section 1.2 and 1.3 to the ten weeks of recovered factor returns above gives the factor covariance matrix (wobbliness and togetherness of the three factors) and, rescaled, the factor correlation matrix (same thing, but always between -1 and +1 so it's easy to read).

Covariance	Market	Value	Moment.
Market	1.999	0.115	0.329
Value	0.115	0.161	-0.074
Momentum	0.329	-0.074	0.232

Correlation	Market	Value	Moment.
Market	1.00	0.20	0.48
Value	0.20	1.00	-0.39
Momentum	0.48	-0.39	1.00

Reading the correlation table: Market and Momentum moved together fairly strongly this sample (correlation 0.48) -- when the market rallied, recently-hot stocks tended to rally even more. Value and Momentum moved somewhat in opposite directions (correlation -0.39) -- weeks that favoured hot/momentum stocks tended to be weeks that were less kind to cheap/value stocks.

3.6 What's left over: each stock's own personality

Subtracting each week's factor-predicted ("fitted") return from the actual return, stock by stock, gives the residual -- exactly Section 1.5's leftover, computed for real:

Week	A	B	C	D	E
1	0.697	0.385	-0.701	-0.718	0.337
2	0.680	0.380	-0.695	-0.766	0.401
3	0.559	0.280	-0.491	-0.152	-0.196
4	-0.002	-0.018	0.044	0.256	-0.281
5	0.991	0.505	-0.890	-0.384	-0.221
6	-0.968	-0.451	0.765	-0.248	0.901
7	-0.647	-0.326	0.573	0.206	0.194
8	0.296	0.124	-0.198	0.291	-0.513
9	-0.059	-0.036	0.067	0.107	-0.079
10	-0.022	-0.036	0.081	0.372	-0.394

Stock	A	B	C	D	E
Specific variance (wk % ²)	0.384	0.099	0.307	0.174	0.190
Specific volatility (wk %)	0.62	0.31	0.55	0.42	0.44

Notice Stock C ("Hot Momentum Co.") has by far the largest specific volatility -- its own private wobble is bigger than everyone else's, on top of whatever the shared factors did. That matches how it was built (Section 3.2): a noisier, more excitable stock.

3.7 Build a portfolio and price its risk

Now hold a real portfolio: 40% Stock B, 35% Stock C, 25% Stock D (0% in A and E). First, find the portfolio's own exposures by taking the weighted average of the holdings' exposures:

```
Portfolio exposure: Market = 1.00 Value = 0.58 Momentum = -0.05
```

Market exposure is exactly 1.00 because the three weights (0.40+0.35+0.25) add up to 100% and every stock has Market exposure 1 -- makes sense, you're fully invested. The portfolio tilts moderately toward Value (mostly from the large weight in cheap Stock B) and is close to flat on Momentum.

Now combine that exposure with the factor covariance matrix (Section 3.5) to get factor risk, and combine the specific variances (Section 3.6) with the squared portfolio weights to get specific risk:

```
Factor variance = 2.162 (per week, in percent-squared) Specific variance = 0.064 Total variance = 2.226
```

Taking the square root (Section 1.2) converts this back into a normal, readable number: weekly volatility of 1.49%, or 10.76% if you annualize it the usual way (multiply the variance by 52 weeks first, then take the square root). Of that total risk, 97.11% comes from the shared factors and only 2.89% from the stocks' own private wobbles -- this portfolio's risk is almost entirely a Market/Value/Momentum story, not a stock-picking story, simply because it's fully invested and reasonably diversified across three names.

3.8 Catching the double-counting trap, live, in our own 5-stock world

Section 2.6 described the trap in words. Here it is with real numbers. Take the pure Momentum factor portfolio -- the specific mix of all five stocks engineered to have exactly 1 unit of Momentum exposure and exactly 0 of Market and Value:

```
Pure Momentum portfolio weights: A=0.211, B=-0.103, C=0.336, D=-0.263, E=-0.182
```

First, the proof that this recipe really works exactly as claimed in Section 2.4 -- not approximately, but exactly: multiplying these weights by the actual stock returns, week by week, reproduces the recovered Momentum factor-return column from Section 3.4 **exactly**, to every decimal place (checked here: True). That is the whole reason double-counting happens -- this portfolio's own observed wobble literally **is** what we call "the Momentum factor's wobble."

```
F_hat (Momentum,Momentum), i.e. the variance of this portfolio's own weekly returns = 0.232
```

Now apply the traditional total-risk formula from Section 2.5 to this same portfolio, the way you would for any portfolio: add a separate specific-risk term on top, computed the normal way from each held stock's specific variance and its weight in this portfolio:

```
naive specific-risk add-on = 0.071 naive total variance = 0.232 + 0.071 = 0.303
```

But the TRUE variance of this portfolio's own return series is just 0.232 -- the exact same number as $F_{\text{hat}}(\text{Momentum}, \text{Momentum})$ above, because (as just proven) they are literally the same return series. The naive formula therefore overstates this portfolio's true risk by 0.071 out of 0.232, an overstatement of **30.70%** -- purely from adding a specific-risk term that was already hiding inside the factor-risk term. With only 5 stocks to spread the private wobble across (even fewer than the 10-stock example in the main workbook), the effect is larger here -- exactly what Section 2.6's "not enough boats" intuition predicts.

Closing Note

Every number in Part 3 was computed from the same 10 weeks of data, following the model's actual mechanics step by step -- nothing was picked to look tidy. If you want to check any of it yourself, the recipe at every stage was: (1) take a wobble, (2) square it or multiply it with another wobble, (3) average over time, (4) take a square root when you

want normal units back. That really is all the statistics a factor risk model needs -- the rest of the KAC3 methodology (term structure, EWMA, PCA shrinkage, the finite-sample adjustment) is just applying these same four ideas more carefully, at greater scale, and with more care about the traps described in Part 2.